

Test worksheet chapter 2

L 001

1. Determine whether the equation $(4x + 1)(3x + 2)/(5x - 3) = x + 1$ has two distinct real roots, two equal roots or no real roots. [4 marks]

2. Find the values of the constant k for which $(2k - 1)x^2 + 6x + k + 1 = 0$ has real roots. [5 marks]

3. Find the possible values of the constant k such that the equation $kx^2 + 4kx + 3k + 1 = 0$ has two different real roots. [4 marks]

4. Write down the values of k for which the equation $(2x+1)^2(x-3) = k$ has exactly one solution. [2 marks]

5. Hence find the values of k for which $f(x) = k$ has exactly one solution. [2 marks]
